

VSL400: A Multi-view Dataset for Vietnamese Word-Level Sign Language Recognition

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Abstract

This research introduces VSL400, a multi-view video dataset created for the isolated word-level recognition of Vietnamese Sign Language (VSL). The dataset comprises 74,259 manually annotated video clips, encompassing 400 glosses, and was performed by 28 signers, including both deaf native users and trained hearing signers. Each sign instance was recorded concurrently from three synchronized RGB camera views (front, left, and right), thereby capturing the inherently three-dimensional character of sign articulation. To facilitate reproducible research, we provide a comprehensive preprocessing pipeline for boundary detection, temporal segmentation, and spatial normalization, along with structured metadata. All video data, annotations, and baseline processing code are publicly accessible under an open license. VSL400 addresses a significant resource deficiency for Vietnamese Sign Language and offers a practical reference dataset for the development and evaluation of robust recognition methods in low-resource environments.

1 Background & Summary

The global public health situation has worsened because 5% of worldwide population now experiences hearing loss which amounts to 430 million people including 34 million children

who require hearing rehabilitation services. The World Health Organization predicts that disabling hearing loss will affect 700 million people during 2050 which will make 10% of the worldwide population experience this condition.¹ According to the World Federation of the Deaf, over 72 million deaf individuals worldwide use more than 300 distinct sign languages for daily communication.^{2,3} Sign language functions as a complete visual language which uses hand gestures and facial expressions and body posture and spatial positioning to transmit meaning and provide essential communication for expressing information and ideas and emotions. Human interpreters who link deaf people to hearing people encounter barriers because their services become too costly for users to access. The situation requires automated sign language technology systems which should function as communication solutions for large-scale applications. The development of sign language recognition (SLR) and sign language translation (SLT) systems creates opportunities to enhance inclusion through immediate language translation services between sign language and spoken or written communication. The systems use machine learning (ML) and artificial intelligence (AI) to convert visual sign data into text or speech and speech into text and speech which enables them to create automatic broadcast captions and interactive system hand movement predictions. These technologies function as essential instruments which help deaf communities to join society and succeed in school and work while providing complete communication access to all people.

Sign languages which deaf people need for their daily existence lack proper representation in standard communication systems which focus on spoken and written language.⁴ Sign languages function as essential communication tools which deaf and hard-of-hearing people need to connect with others and learn at school and maintain their cultural heritage.⁵ People who have hearing impairments encounter major difficulties because digital systems and communication systems do not have built-in functionality to support their visual-gestural communication needs. Sign languages across different countries use separate systems which create additional challenges because each system has its own grammatical structure and word selection and sentence organization rules. The United States uses American Sign Language (ASL) as its sign language system while China uses Chinese Sign Language (CSL) and other countries have their own sign languages including Arabic Sign Language and German Sign Language and Indian Sign Language and Korean Sign Language.⁶⁻¹¹

The solution of these problems requires technologically inclusive answers because they will both improve accessibility and create linguistic equality which supports social inclusion for deaf people in various cultural settings worldwide.

1.1 Available Sign Language Datasets

The progress of sign language recognition (SLR) research has been substantially driven by the availability of curated datasets. However, the development of robust and generalizable models remains constrained by the limited size, modality coverage, and linguistic diversity of existing resources particularly in underrepresented sign languages.

In the domain of isolated sign language recognition (ISLR), several datasets have gained prominence: BSL-1K¹² is a large-scale British Sign Language dataset comprising over 1,000 hours of broadcast footage, weakly annotated with 1,000 glosses through self-supervised mining techniques. While rich in content, the dataset is weakly supervised and lacks temporal

segmentation. The How2Sign dataset¹³ provides over 80 hours of synchronized multi-view and multimodal recordings of American Sign Language (ASL), including RGB video, depth data, skeletal keypoints, and text transcripts.

For continuous SLR, RWTH-Phoenix-Weather 2014¹⁴ remains a widely adopted benchmark. This German Sign Language dataset contains 5,672 training, 540 validation, and 629 test videos, recorded at 25 FPS by nine signers, with a vocabulary of 1,295 glosses. Despite its popularity, it is domain-specific and lacks visual diversity. Other datasets include AUTSL,¹⁵ with 38,000 Turkish Sign Language videos across 226 classes, and CSL-Daily,¹⁶ a large-scale Chinese Sign Language corpus with gloss and textual annotations. The BBC-Oxford BOBSL dataset,¹⁷ while high in quality and produced by professional signers, remains limited in scale relative to the expressive complexity of natural sign languages.

Table 1 summarizes these datasets and highlights key attributes relevant to benchmark development. Notably, Vietnamese Sign Language (VSL) remains absent from public datasets, motivating the construction of VSL400, a multi-view, high-quality resource designed for reproducible word-level SLR in a low-resource setting.

Table 1: Comparison of benchmark sign language datasets.

Dataset	Lang	Glosses	Clips / Hours	Signers	Modality	View
BSL-1K	BSL	~1,000	~1,000 hrs	N/A	RGB (TV)	Frontal
How2Sign	ASL	~16,000	80 hrs	~10	RGB, Depth, Pose	Multi
RWTH-Phoenix14T	DGS	1,295	7,000+ videos	9	RGB	Frontal
AUTSL	TSL	226	38,000 videos	43	RGB, Depth	Frontal
CSL-Daily	CSL	~500	50,000+ videos	50	RGB, Text	Frontal
BOBSL	BSL	~1,000	~1,400 hrs	N/A	RGB	Frontal
VSL400 (Ours)	VSL	400	74,259 videos	28	RGB	Multi

1.2 Automatic Sign Language Recognition

Multiple machine learning methods have been studied for automatic sign language recognition (SLR) during the last few decades. The SLR pipelines from before the deep learning period used manually designed visual features together with Hidden Markov Models (HMMs) and Conditional Random Fields (CRFs) for sequential modeling of time-dependent data.^{18–20} The classification system used Support Vector Machines (SVMs) as classifiers to convert engineered features into pre-defined gloss labels.

The field has experienced a fundamental change through deep learning technology which emerged during the last ten years because researchers now use end-to-end trainable frameworks that combine CNNs for video feature extraction with RNNs and attention-based mod-

els for temporal pattern analysis. The new approach has brought about major enhancements in recognition system accuracy when tested on different evaluation standards.

Contemporary deep learning-based SLR frameworks can be broadly categorized into two principal families: RGB-based (visual) and skeleton-based approaches.

The RGB-based SLR system creates detailed models through visual data processing of RGB video streams which merge spatial and temporal information. The first models C3D²¹ and Inflated 3D ConvNets (I3D)²² used 3D convolutional kernels to detect motion patterns. The models achieved excellent results during training with big datasets which included MS-ASL²³ and WLASL.²⁴ Research has added sign-specific inductive biases to its methods through hand-pose-guided 3D pooling which according to Hosain (2021) focuses on hand and facial areas to boost discrimination power.

More recently, transformer-based architectures have achieved state-of-the-art performance in video understanding. Video Swin Transformer²⁶ employs a hierarchical shifted-window attention mechanism to capture both local and global dependencies. VideoMAE²⁷ introduces masked video modeling with high masking ratios, enabling self-supervised learning from unlabeled video through spatiotemporal reconstruction.

Skeleton-based SLR represents the signer using landmark coordinates extracted from RGB frames, thereby reducing background noise and computational overhead. Popular pose estimation models include OpenPose,²⁸ HRNet,²⁹ and BlazePose.³⁰ Given sequences of skeletal joints, Spatial-Temporal Graph Convolutional Networks (ST-GCN)³¹ model these as graphs with both spatial and temporal edges. MS-G3D³² extends this by introducing multi-scale graph convolutions and gated G3D operators to capture long-range dependencies.

Transformer-based skeleton models, such as SPOTER,³³ apply self-attention to landmark sequences, while hybrid models integrate GCNs with temporal transformers.³⁴

In multi-modal systems, SAM-SLR³⁵ fuses RGB, depth, and skeletal modalities using sign-specific SL-GCN and SSTCN components. DSTA-SLR³⁶ further enhances recognition robustness by dynamically aggregating spatial-temporal relations and incorporating multi-scale temporal representations. These approaches achieve improved generalization and robustness, particularly on large-scale benchmarks such as WLASL.²⁴

1.3 Vietnamese Sign Language

Vietnamese Sign Language (VSL) presents a number of unique linguistic characteristics that differentiate it from many other signed languages, notably its pervasive use of ellipticity and focus-driven emphasis. The former involves the systematic omission of pragmatically recoverable components such as null arguments, semantically lightweight function words, and auxiliaries-made possible by VSL’s highly developed spatial indexing and classifier mechanisms. The latter encompasses the use of non-manual signals such as eyebrow movement, head tilts, and mouthing, prosodic features such as holds and segmental lengthening, and flexible constituent ordering to convey discourse-level emphasis. These features result in a discourse-adaptive syntax in which functional lexical elements are often omitted unless they contribute substantially to propositional meaning.

In Vietnam, there are an estimated 2.5 million deaf or hard-of-hearing individuals who rely on VSL for everyday communication. Despite the scale and importance of this user community, technological resources tailored to VSL remain scarce. The absence of reliable

sign language recognition (SLR) tools exacerbates communication inequities and hinders full participation in education, employment, and public life.

While the linguistic and cultural distinctiveness of VSL has been acknowledged, it remains largely underrepresented in computational research, particularly within data-driven approaches. Existing efforts to develop automatic VSL recognition systems^{37–39} have been constrained by the lack of publicly available, large-scale, and high-quality datasets. This underrepresentation limits not only model performance and generalization, but also the broader inclusion of VSL in international research on sign language technologies.

1.4 Contributions of This Study

Sign language recognition (SLR) approaches are generally categorized into two primary types: word-level recognition,^{40,41} which focuses on identifying individual sign glosses, and sentence-level recognition,^{42–44} which aims to interpret continuous sign sequences. This study specifically addresses the former, as word-level recognition forms a critical foundation for developing more complex continuous and translation-level models.

Research has expanded to develop and evaluate word-level datasets but most studies have used recordings from the front view. The natural three-dimensional structure of sign language together with its dependence on context makes it difficult to understand when viewers see the signs from different positions. The training process which uses only frontal-view data creates performance limitations for models when they need to function in various situations. The front view of multiple glosses shows identical visual patterns which makes it harder to distinguish between different classes. The solution requires us to create a dataset which shows each sign through three different views that include frontal and left and right perspectives to achieve better simulation of actual usage and enhance system stability.

This study introduces VSL400, the first publicly available benchmark dataset for word-level Vietnamese Sign Language (VSL) recognition, designed to support reproducible and data-driven research in a low-resource sign language setting. The main contributions of this work are summarized as follows:

First, we present a large-scale, manually annotated word-level VSL dataset comprising 74,259 video clips covering 400 glosses, recorded from 28 signers, including deaf native users and trained hearing signers. The dataset addresses the absence of publicly accessible resources for Vietnamese Sign Language and provides a foundational corpus for isolated sign recognition research.

Second, the dataset is collected using a synchronized tri-view recording protocol, with frontal, left, and right camera perspectives captured simultaneously for each sign instance. This design reflects the inherently three-dimensional nature of sign language articulation and enables systematic investigation of cross-view variability, viewpoint robustness, and multi-view learning strategies.

Third, we introduce a reproducible preprocessing and segmentation pipeline that converts continuous multi-view recordings into standardized, gloss-level video clips. The pipeline combines pose-based temporal boundary localization with boundary-guided segmentation and spatial normalization, and is released alongside the dataset to facilitate transparent reuse and fair benchmarking.

Fourth, we provide structured metadata, standardized data organization, and comprehensive quality control procedures, including multi-pass manual annotation, inter-annotator agreement verification, and cross-view synchronization checks. These measures ensure the internal consistency, reliability, and long-term usability of the released data.

Finally, we report reference baseline results using five representative deep learning models spanning both RGB-based and skeleton-based recognition paradigms. These experiments are intended to serve as technical validation of the dataset and as reproducible reference points, rather than exhaustive performance comparisons, thereby supporting future methodological research built upon VSL400.

2 Materials and Methods

This section outlines the procedures used for data acquisition, preprocessing, annotation, and segmentation of video recordings into structured gloss-level units. The complete dataset collection and processing pipeline is illustrated in Figure 1.

To construct the dataset, tri-view continuous video recordings of sign sequences were captured using synchronized RGB cameras placed at frontal, left, and right viewpoints. Each session involved a signer performing a predefined set of glosses in a controlled indoor environment with consistent background and lighting conditions.

The raw video files were then subjected to a normalization process to ensure temporal alignment across the three views and standardize encoding parameters such as frame rate, resolution. Following synchronization, automatic segmentation was performed using keypoint-based boundary detection with the Mediapipe framework. This process enabled identification of sign boundaries and extraction of individual word-level clips.

Subsequently, a semi-automatic annotation procedure was employed. Candidate clips were imported into the Label Studio platform, where trained annotators manually verified gloss labels, ensured segmentation quality, and corrected any misalignments or mislabeled instances. The resulting dataset includes high-quality, manually validated samples suitable for reproducible training and benchmarking of sign language recognition systems.



Figure 1: Dataset collection pipeline.

2.1 Participant Recruitment and Data Collection Protocol

We used video recordings from 28 participants to create the dataset which included two separate participant groups. The first group included 23 deaf students recruited from specialized education centers for the hearing-impaired. These participants are native users of Vietnamese Sign Language (VSL) and contributed the majority of the dataset. The studio

recording area used standardized lighting conditions and fixed camera positions and background setup to create high-quality video while excluding all external elements which could affect the results.

The dataset needs more diverse visual content which includes different speaking methods and various settings to achieve better results. The researchers recorded five additional hearing participants inside a classroom setting which provided them with less controlled recording conditions. The training process for these people involved them to create VSL glosses through the same recording method while deaf experts provided them with assistance. The addition of this group segment brings average diversity to both signer profiles and recording environments which makes the dataset more suitable for real-world modeling applications.

All participants performed a predefined vocabulary of 400 glosses, recorded from three synchronized camera views (frontal, left, and right) at a resolution of 1920×1080 pixels and 30 frames per second. The participants received instructions to look directly at the cameras while keeping their body position stable throughout the signing process for better cross-view alignment and pose estimation.

2.2 Gloss Selection and Vocabulary Construction

The VSL400 vocabulary was designed to reflect high frequency and semantically diverse concepts used in daily communication. Glosses were selected and reviewed by expert annotators fluent in VSL and knowledgeable in linguistic annotation. The vocabulary spans four semantic domains: People and Society, Daily Life, Nature and Environment, and Activities and Concepts. This structured approach supports future domain-specific evaluations and curriculum-based modeling.

A core innovation of our data collection protocol is a three-camera configuration with sensors positioned on the left ($\sim 45^\circ$), Front ($\sim 0^\circ$), and Right ($\sim 45^\circ$) as shown in Fig. 2. This arrangement, comparatively uncommon in existing word-level Sign Language Recognition (SLR) datasets, captures complementary multi-view observations of each sign, thereby enriching the dataset. The resulting diversity of viewpoints enhances its utility for machine learning and computer vision research, enabling more accurate and robust recognition models.

2.3 Video Preprocessing and Segmentation

To convert raw continuous video recordings into structured and normalized gloss-level clips suitable for training, annotation, and evaluation, we designed a fully automated two-stage pipeline composed of: (1) **Temporal Boundary Localization (TBL)** and (2) **Boundary-Guided Segmentation and Pruning (BGSP)**. The processing steps are visualized in Figure 2, and the corresponding pseudocode is provided in Algorithm 1 and Algorithm 2.

In the first stage, TBL (Algorithm 1) detects candidate gesture segments by analyzing elbow joint dynamics. Using MediaPipe, we extract the 2D coordinates and visibility scores of left/right shoulder, elbow, and wrist landmarks across video frames. Elbow angles are computed and binarized using a threshold ($\theta = 160^\circ$), followed by a temporal smoothing



Figure 2: Tri-camera VSL data acquisition pipeline: synchronized left–front–right views, fixed camera geometry, controlled lighting, and standardized signer enrollment.

mechanism that enforces gesture persistence over N consecutive frames. Transitions from inactive to active and back are used to define the start and end of candidate intervals, with symmetric buffer padding (τ_b) to capture gesture context.

In the second stage, BGSP (Algorithm 2) refines and transforms the candidate segments into final gloss-level video clips. Each segment is checked for minimum duration ($\tau_{min} = 0.67s$) and optionally merged with nearby segments. We then extract frames from the corresponding timestamp range and crop them spatially to a fixed 1080×1080 resolution centered on the signer, using an empirically optimized horizontal offset ($x_{offset} = 420$). The resulting clips are stored in a normalized format and indexed by gloss labels for further annotation and model training.

The system provides a method to perform segmentation of sign language data which produces results that are both reproducible and have consistent resolution and semantic accuracy for model training and benchmark evaluation.

Algorithm 1 performs Temporal Boundary Localization (TBL) to extract candidate gesture intervals based on arm joint angles computed from pose landmarks.

Algorithm 1: Temporal Boundary Localization (TBL).

```
Input:  $F = \{f_1, f_2, \dots, f_T\}$  // video frames
1  $fps$  // frame rate (e.g., 30 FPS)
2  $\theta = 160^\circ$  // elbow angle threshold for inactivity
3  $\tau_p = N/fps$  // persistence threshold (seconds)
4  $\tau_b$  // symmetric temporal buffer (e.g., 0.4s)
Output:  $S$  = list of candidate active segments  $[start\_time, end\_time]$ 
5  $S \leftarrow \emptyset$ ; // initialize output segment list
6 foreach frame  $f_i$  in  $F$  do
7   Use MediaPipe to extract landmarks: left/right shoulder, elbow, wrist;
8   if all 3 landmarks have visibility  $\geq 0.6$  then
9     Compute elbow angle  $\alpha_i$ ;
10    if  $\alpha_i < \theta$  then
11      Set arm state  $s_i = 1$  (active);
12    else
13      Set  $s_i = 0$  (inactive);
14  else
15    Set  $s_i = 0$ ; // assume inactive due to occlusion
16 Apply temporal smoothing: enforce persistence of  $N$  frames;
17 Only mark transition if active/inactive state lasts  $\geq N$  frames;
18 foreach transition  $[t_{start}, t_{end}]$  in  $s$  do
19    $start\_time \leftarrow \max(0, t_{start}/fps - \tau_b)$ ;
20    $end\_time \leftarrow \min(T/fps, t_{end}/fps + \tau_b)$ ;
21   Append  $[start\_time, end\_time]$  to  $S$ ;
22 return  $S$ 
```

Algorithm 2 applies Boundary-Guided Segmentation and Pruning (BGSP), which merges noisy intervals, discards short gestures, and performs spatial cropping to generate normalized 1080×1080 clips.

Algorithm 2: Boundary-Guided Segmentation and Pruning (BGSP).

```
Input:  $S$  // List of candidate segments from Algorithm 1
1  $H = 1080, W = 1920$  // Original video resolution
2  $crop\_size = 1080 \times 1080$ 
3  $x_{offset} = 420$  // Horizontal crop offset
4  $\tau_{min} = 0.67$  seconds // Minimum gesture duration
Output:  $Clips$  = List of final processed video clips
5  $Clips \leftarrow \emptyset$ ;
6 foreach  $segment [start\_time, end\_time] \in S$  do
7   if  $end\_time - start\_time < \tau_{min}$  then
8     continue ; // Discard short segments
9   if  $overlap\ with\ previous\ segment < gap\_threshold$  then
10    Merge segments;
11     $start\_frame \leftarrow \lfloor start\_time \times fps \rfloor$ ;
12     $end\_frame \leftarrow \lceil end\_time \times fps \rceil$ ;
13     $Buffer \leftarrow \emptyset$ ;
14    for  $f_i$  in  $[start\_frame, end\_frame]$  do
15      Crop  $f_i$  at:  $x = x_{offset}, y = 0, width = height = 1080$ ;
16      Append cropped frame to  $Buffer$ ;
17     $Clip \leftarrow$  Compile  $Buffer$  to video;
18    Assign unique ID and gloss label to  $Clip$ ;
19    Save  $Clip$  to output directory;
20    Append  $Clip$  to  $Clips$ ;
21 return  $Clips$ 
```

All segmented clips were saved with consistent naming conventions, unique identifiers, and metadata files specifying gloss label, signer ID, and recording parameters. Both algorithms are implemented in Python using OpenCV and MediaPipe. The code and configuration scripts are publicly available (see Code Availability section).

2.4 Annotation and Quality Control

Gloss annotations were manually verified by trained sign linguists through a two-pass validation process. Each clip was reviewed for signing accuracy, gloss alignment, and visual clarity. Annotators were provided with recording protocols and domain, specific gloss lists to ensure consistency. Discrepant cases were adjudicated jointly to reach consensus. Annotation consistency exceeded 95% agreement across the corpus.

2.5 Dataset Organization

The final dataset consists of 74,259 annotated video clips distributed across 400 gloss labels. Each gloss includes multiple samples from different signers and from all three camera views. The dataset is organized into three main folders-`front_view`, `left_view`, and `right_view`

each containing synchronized video clips with shared six-digit identifiers across views. Associated JSON metadata files encode gloss labels, signer ID, frame indices, and timing data for each clip.

All code for preprocessing, segmentation, and annotation validation is publicly available to support reproducibility and community-driven extensions.

Table 2 summarizes key dataset statistics. Data were collected from 28 signers of varying age and gender, with each gloss captured simultaneously from three viewpoints (front_view, left_view, right_view) to support multi-view modeling. After segmentation and normalization, the dataset spans approximately 53.99 hours of video, with an average clip duration of 2.61 seconds.

The scale, diversity, and structure of VSL400 offer a valuable benchmark for word-level sign language recognition, signer-invariant modeling, and low-resource learning in underrepresented sign languages.

Table 2: Statistics of the VSL400 Dataset.

Attribute	Value	Notes
Number of glosses	400	Annotated manually
Total videos	74,259	Across all views
Average videos per gloss	185.6	Balanced class distribution
Number of signers	28	Diverse age and gender
Recording views	3	Frontal, Left, Right
Clip duration (avg)	2.61 s	After boundary pruning
Total duration	53.99 hours	Normalized format

2.6 Dataset Statistics and Distribution Analysis

Figures 3–10 provide a comprehensive statistical overview of the VSL400 dataset, characterizing its vocabulary composition, temporal properties, signer contributions, and sample balance. Together, these analyses offer insight into the internal structure and reliability of the dataset, as well as its suitability for isolated sign language recognition research.

Figure 3 illustrates the distribution of vocabulary groups within VSL400. The dataset covers a broad range of semantic categories, with particular emphasis on everyday concepts and environmental themes that are commonly encountered in daily communication. This distribution reflects a design choice to prioritize general-purpose vocabulary rather than domain-specific terminology, thereby increasing the dataset’s applicability to real-world sign language recognition scenarios.

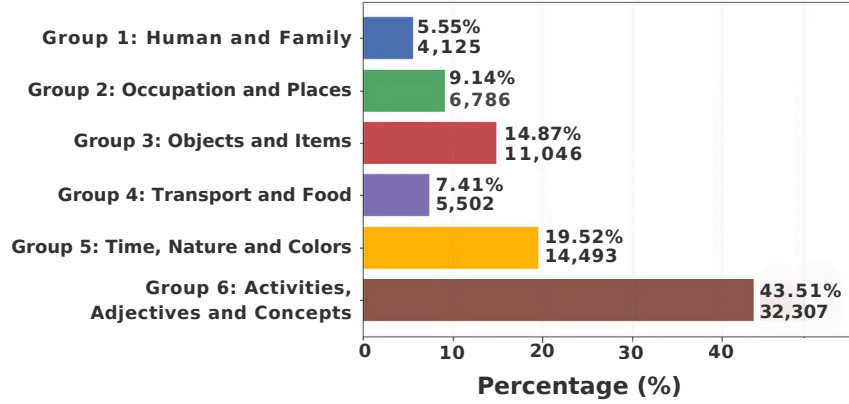


Figure 3: Percentage and total numbers of each vocabulary group.

The frequency distribution of individual glosses is further examined in Figure 4, which shows the top ten most frequent signs by sample count. While certain commonly used glosses appear more frequently, no single gloss dominates the dataset. This controlled variation mirrors natural language usage patterns while avoiding extreme class imbalance that could bias model training or evaluation.

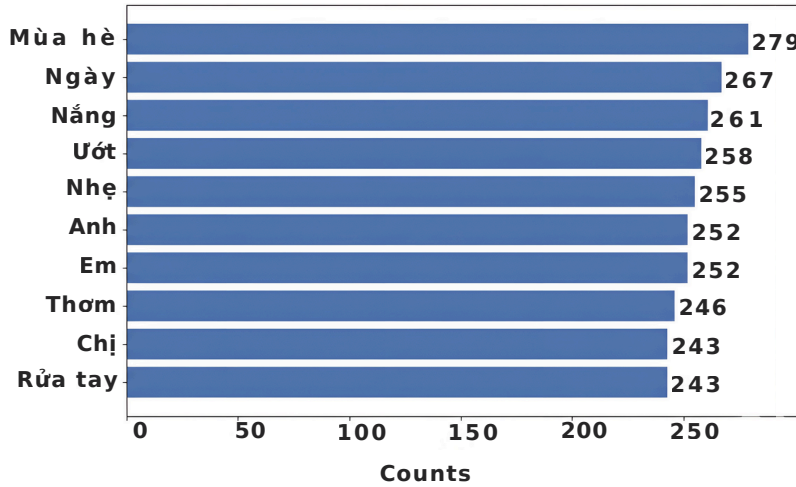


Figure 4: Top 10 signs by count.

Temporal characteristics of the dataset are summarized in Figures 5 and 6. Figure 5 presents the overall distribution of clip durations across all samples, revealing a tightly clustered range with closely aligned mean and median values. This indicates consistent temporal segmentation and suggests that redundant pre- or post-sign motions have been largely excluded during preprocessing. Figure 6 complements this analysis by showing the total accumulated duration per gloss, highlighting moderate variability that is expected due to differences in articulation complexity across signs.

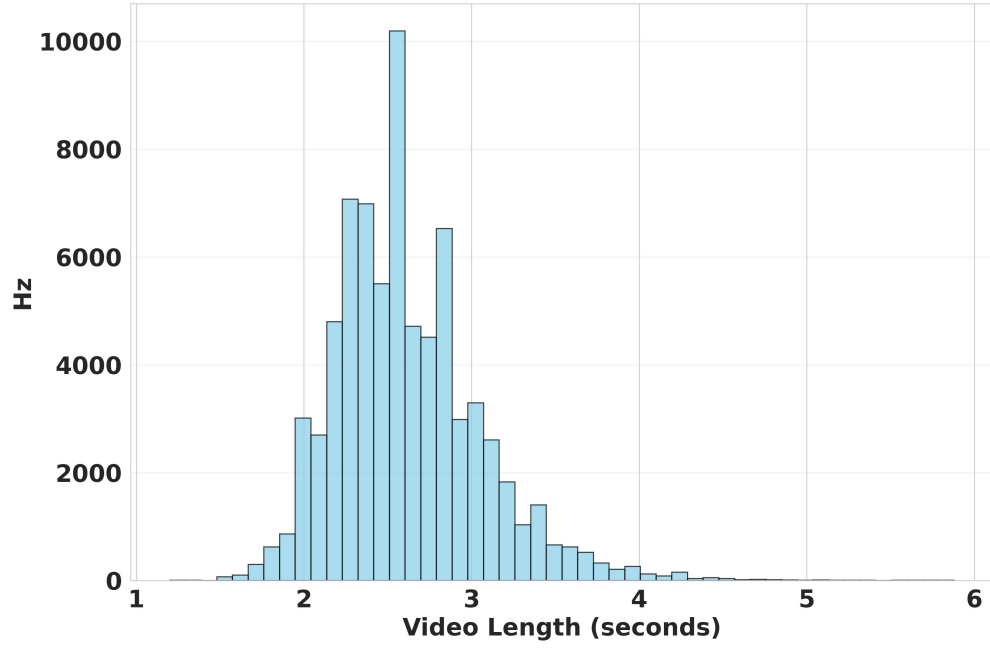


Figure 5: Distribution of video length on all samples.

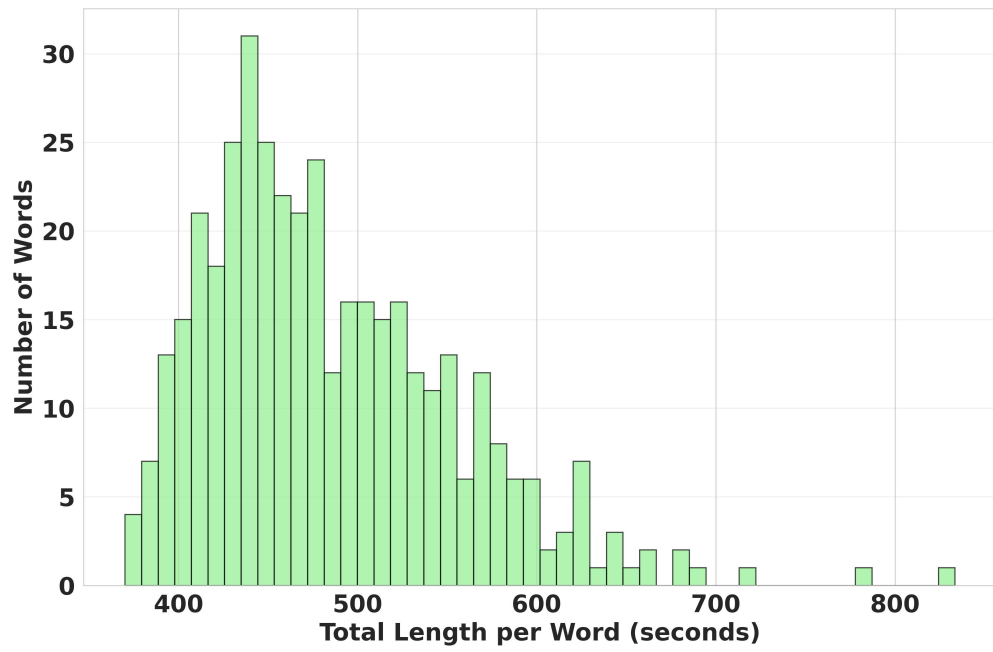


Figure 6: Distribution of total length for each signs.

Figures 7 and 8 further explore duration-related properties at a finer granularity. Figure 7 reports the top twenty glosses by total video length, reflecting signs that are either frequently articulated, temporally longer, or both. Figure 8 illustrates the range of clip durations for high-frequency glosses, revealing noticeable intra-class variability. Such variation arises from differences in signing tempo, emphasis, and individual articulation styles, and underscores the importance of temporal robustness in recognition models trained on the dataset.

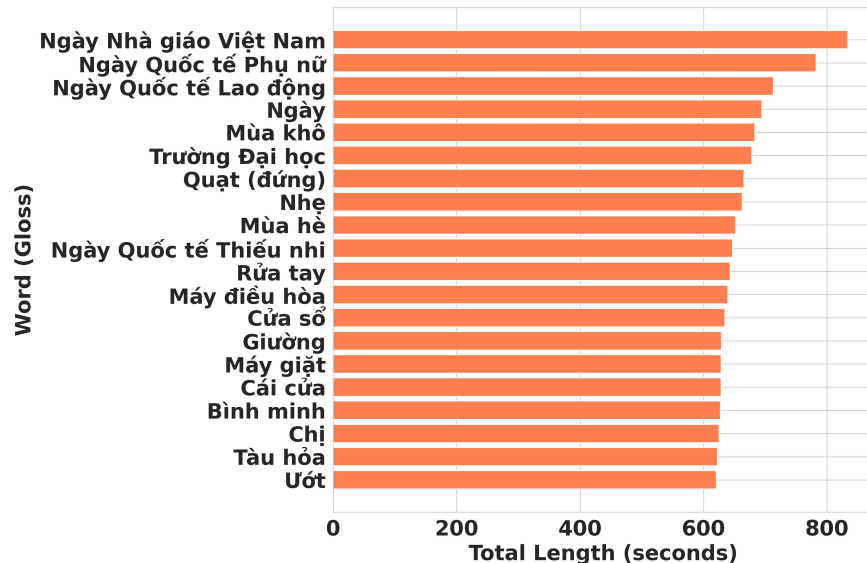


Figure 7: Top 20 words by total length.

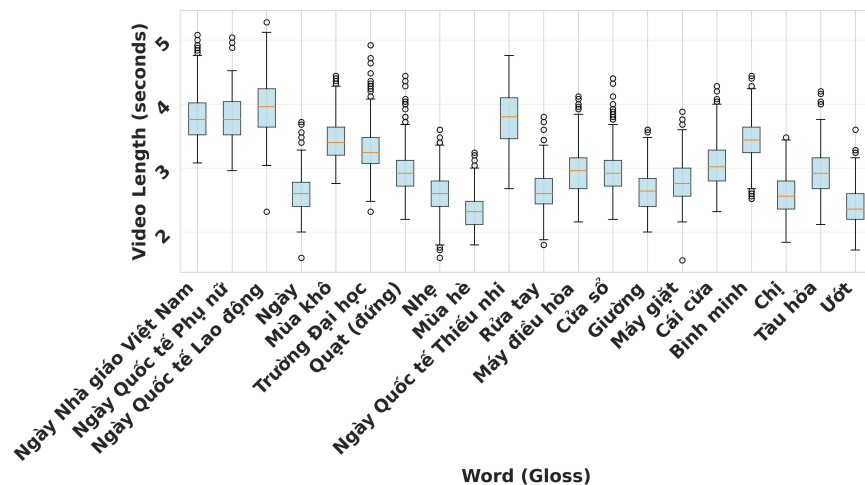


Figure 8: Length range of top 15 signs.

Signer diversity and contribution balance are examined in Figure 9, which displays the distribution of samples across the top contributing signers. Although individual signers contribute different numbers of samples, no single signer disproportionately dominates the dataset. This balance supports the use of signer-disjoint evaluation protocols and reduces the risk of models overfitting to signer-specific appearance or motion patterns.

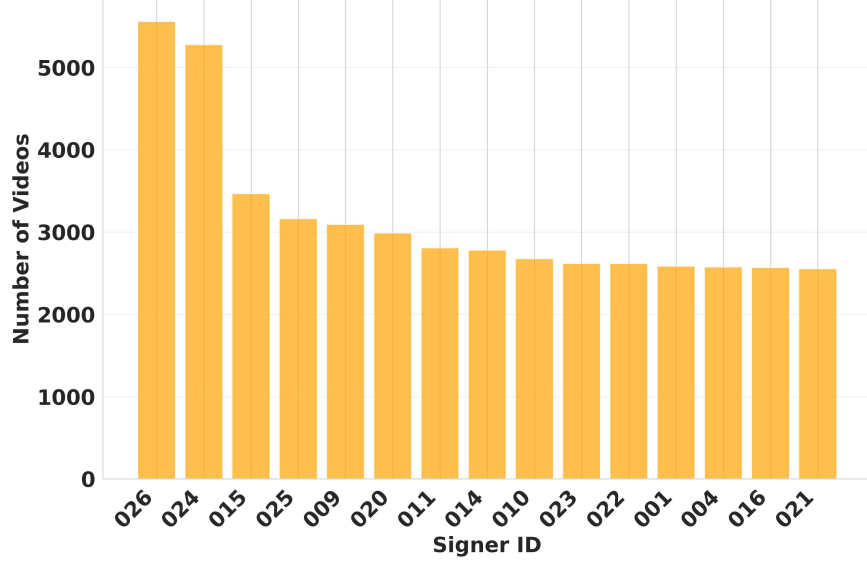


Figure 9: Top 15 signers by sample count.

Figure 10 presents a summary of the sample distribution for each gloss throughout the complete vocabulary. The majority of signs are associated with a similar number of samples; the observed moderate variation is attributable to the realities of data collection, rather than any inherent bias. This relatively equitable distribution promotes consistent training performance and enables unbiased assessment across different gloss categories.

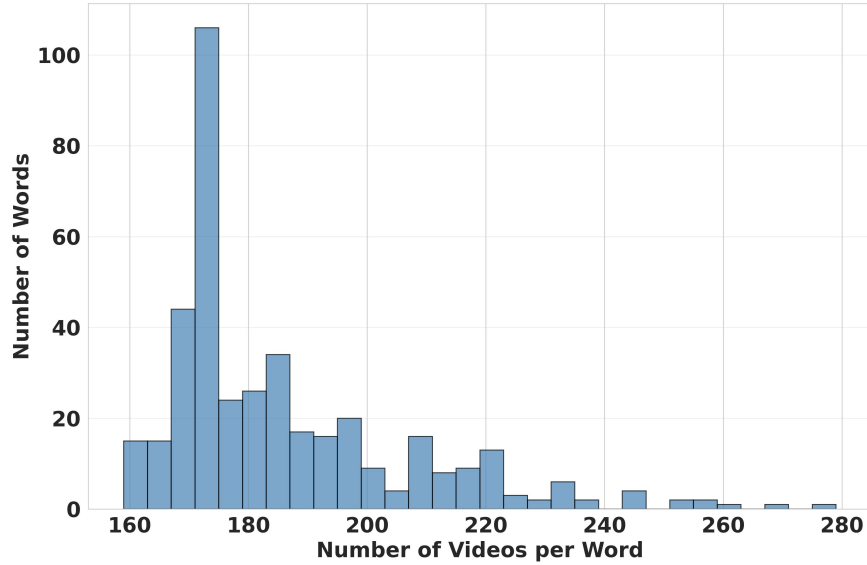


Figure 10: Distribution of signs by number of samples per sign.

The statistical analyses in Figures 3–10 indicate that VSL400 exhibits balanced vocabulary coverage, consistent temporal segmentation, and well-distributed signer contributions. These properties demonstrate the structural reliability of the dataset and support its use as a practical reference for isolated Vietnamese Sign Language recognition. In addition,

the observed distributions reflect realistic variability in signing styles and durations without introducing extreme imbalance, thereby facilitating stable training and fair evaluation. Together, these characteristics make VSL400 well suited for studies on robustness, signer generalization, and multi-view learning.

Figures 11 and 12 illustrate the visual diversity and structural robustness of the VSL400 dataset. Figure 11 shows instances of the same gloss recorded simultaneously from three camera viewpoints, demonstrating consistent multi-view acquisition. Figure 12 presents different signers performing the same gloss under varied appearance conditions such as clothing styles, underscoring signer and visual variability that supports model generalization across individuals and contexts.

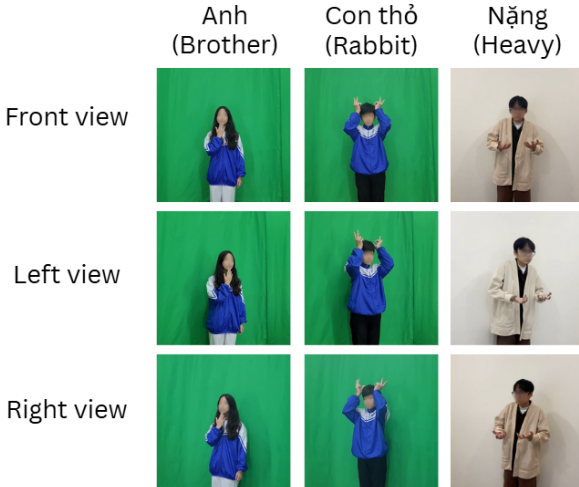


Figure 11: Same gloss sample from different camera view.



Figure 12: Different signers and outfits for same gloss.

2.7 Data availability

Our dataset is available online at VSL400 dataset. As illustrated in Figure 13, the archive contains three camera folders (`front_view`, `left_view`, `right_view`) with synchronized multi-view recordings of sign language glosses, where videos with matching six-digit IDs across folders represent frontal, left, and right views of the same sign. Metadata files (`*_view.json`) provide gloss labels and recording specifications for each camera view. Figure 14 shows the metadata for each file. This includes the video ID, signer ID, gloss label, frame count, and runtime for every clip.

To protect participant privacy, all video data have undergone facial blurring using automated computer vision filters. This anonymization ensures that signer identities cannot be inferred during public use or model training.

All videos are normalized in resolution and frame rate to support standardized benchmarking.

🕒 Recents ☆ Starred

Name ↑

<code></></code>	front_view.json
<code></></code>	left_view.json
<code></></code>	right_view.json
📦	VSL400_blurred.rar

```

1 [
2   {
3     "video_id": "000000",
4     "signer_id": "001",
5     "fps": 25.0,
6     "resolution": 1080,
7     "gloss": "Anh",
8     "num_frames": 67,
9     "length_seconds": 2.68
10  },
11  {
12    "video_id": "000001",
13    "signer_id": "001",
14    "fps": 25.0,
15    "resolution": 1080,
16    "gloss": "Anh",
17    "num_frames": 58,
18    "length_seconds": 2.32
19  },

```

Figure 13: File structure of the VSL400 distribution.

Figure 14: Example of a JSON metadata entry.

3 Technical Validation

To ensure the reliability and quality of the VSL400 dataset, we implemented a rigorous multi-stage validation process encompassing both data integrity verification and systematic performance benchmarking. The validation procedure consisted of two primary components: (1) comprehensive quality control checks to verify annotation accuracy and temporal consistency across multi-view recordings, and (2) extensive benchmarking experiments using state-of-the-art models in both RGB and pose-based sign language recognition. These evaluations not only quantify baseline model performance under controlled protocols but also serve as empirical evidence of the dataset’s technical robustness and utility for advancing sign language recognition research.

3.1 Data Quality Assurance

Prior to model evaluation, we conducted a systematic two-stage quality control process to validate all annotations and ensure temporal consistency across the multi-view recordings.

3.1.1 Length Verification

To identify potential annotation errors and redundant content, we performed statistical analysis of video durations across all samples. Violin plots were generated for each sign class to visualize the distribution of video lengths and detect outliers-videos that were unusually short or excessively long compared to the class median. Videos flagged as outliers underwent manual re-examination to determine whether they contained redundant actions, incomplete gestures, or unnecessary pre/post-sign movements. Videos confirmed to have redundant content were precisely trimmed to retain only the core sign execution, ensuring temporal consistency across the dataset. The violin plot, as shown in Fig. 15, highlights the distribution of video lengths for a representative sign class. The kernel density estimate shows where the samples are concentrated around the median length, while the individual dots emphasize probable outliers.

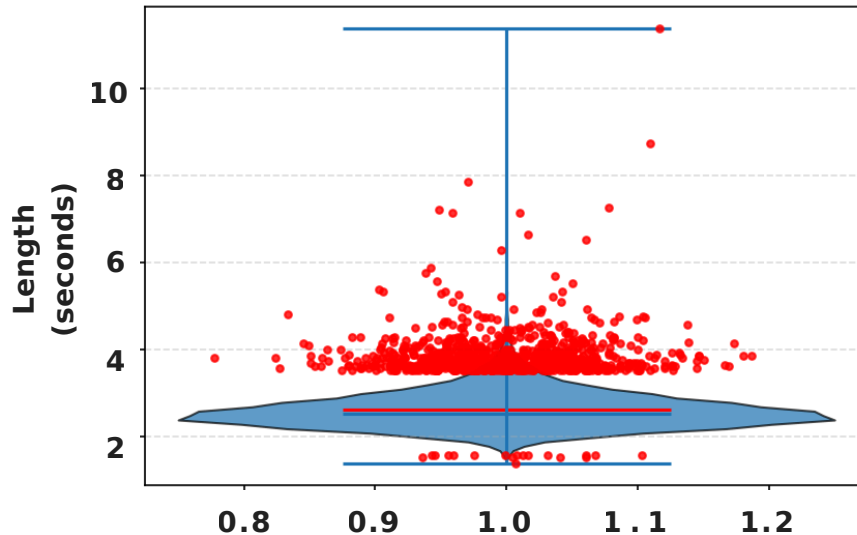


Figure 15: Violin plot of video length distribution for duration-based validation.

3.1.2 Multi-View Synchronization Verification

To verify annotation accuracy and temporal alignment across viewpoints, we implemented a synchronized verification protocol. For each sign instance, the three corresponding videos (front, left, and right views) were displayed simultaneously alongside their annotated labels in a custom verification interface. As illustrated in Fig. 16, this interface allowed human validators to assess each sign from multiple perspectives in real time. Human reviewers checked that: (1) all three videos captured the same sign performance, (2) the temporal alignment across views was maintained, (3) the assigned labels were correct for all viewpoints, and (4) no view-specific recording artifacts or synchronization drift existed. Any discrepancies identified during this process triggered re-annotation or re-recording of the affected samples to maintain dataset integrity.



Figure 16: Example of simultaneous cross-view validation.

3.2 Model Selection and Implementation

For technical validation of the VSL400 dataset, we report reference baselines using five representative deep learning models that cover the major paradigms commonly adopted in vision-based sign language recognition. The objective of this evaluation is not to exhaustively optimize recognition performance, but rather to provide reproducible reference points that help assess dataset consistency and illustrate the impact of the proposed tri-view recording design across different modeling assumptions.

- **RGB-based model (VideoMAE):**²⁷ a masked autoencoder for video that operates directly on spatiotemporal RGB input and has demonstrated strong performance on large-scale SLR and action-recognition benchmarks. In our setting, VideoMAE is fine-tuned on VSL400 clips using frames from the front view or from all three synchronized views.
- **RGB-based model (Swin3D):**⁴⁵ a transformer backbone designed for spatiotemporal visual understanding, employing hierarchical attention to capture both local motion cues and long-range dependencies in video sequences. Swin3D is included as an additional RGB reference model to examine whether the benefits of multi-view data generalize across transformer-based video architectures.
- **Multimodal model (SAM-SLR):**³⁵ a skeleton-aware multimodal recognition framework that integrates complementary visual appearance and pose-based cues through modality-specific encoders and a fusion mechanism. This model is used to evaluate how synchronized multi-view recordings support multimodal feature learning.
- **Skeleton-aware model (DSTA-SLR):**³⁶ a skeleton-aware approach that emphasizes dynamic spatial-temporal aggregation to better capture articulation patterns and temporal relations in sign language sequences. On VSL400, DSTA-SLR serves as a representative skeleton-aware baseline to assess the robustness of structured motion modeling under different camera viewpoints.

- **Skeleton-based model (SPOTER):**³³ a sign pose-based transformer that consumes sequences of 2D body landmarks and is designed to be efficient and robust to signer variation. On VSL400, SPOTER is trained on pose sequences extracted independently from each camera view and then evaluated in both single-view and tri-view regimes.

Across all baselines, we adopt the respective official implementations with minimal modifications and apply a unified evaluation protocol, including signer-disjoint data splits, consistent clip normalization, and matched training and evaluation settings. This design choice ensures that the reported results primarily reflect the intrinsic properties of the VSL400 dataset and the impact of the tri-view setup, rather than architecture-specific optimization.

3.3 Cross-View Evaluation

Table 3: Baseline results for technical validation of the VSL400 dataset under front-view and tri-view training configurations.

Model	Front-view (Top-1 / Top-5)	Tri-view (Top-1 / Top-5)	Gain (Top-1 / Top-5)
VideoMAE	76.28 / 90.18	80.62 / 94.26	+4.34 / +4.08
Swin3D	90.03 / 95.68	93.79 / 96.35	+3.76 / +0.67
DSTA-SLR	90.74 / 95.72	92.85 / 96.13	+2.11 / +0.41
SAM-SLR	91.05 / 97.42	94.84 / 96.62	+3.79 / +1.78
SPOTER	93.58 / 96.62	96.19 / 97.58	+2.61 / +0.96

To examine the effect of multi-view recordings on dataset consistency and recognition robustness, we evaluate all reference models under both single-view (front-view only) and tri-view training configurations. The dataset contains three parts which consist of training data (80%) and validation data (10%) and test data (10%)

The evaluation of isolated sign language recognition uses Top-1 and Top-5 accuracy as standard metrics which offer different views about how well the system recognizes signs. The research evaluates how multi-view recordings affect recognition robustness through these performance metrics which impact different modeling approaches.

The results from Table 3 show that all tested models achieve better recognition results when they receive training data from three views instead of using front-view images alone. The trend appears in all three types of architectures which include RGB-based and skeleton-based and multimodal systems because the additional views provide extra visual and motion information which goes beyond what a single frontal view can offer.

The research shows that the achieved performance enhancements affect all model types without any particular preference. RGB-based architectures achieve better spatial coverage and lower viewpoint ambiguity through their design while pose- and skeleton-aware models extract landmarks more accurately because they minimize self-occlusion when viewing subjects from angles other than straight on. The research results show that the tri-view design provides better stability for data-level sign representation without showing any preference for particular architectural structures.

The assessment results confirm all design principles of VSL400 because it shows that multiple camera views produce dependable and organized performance improvements for different recognition systems. The research results validate the data construction method which demonstrates the need to collect sign language information through various viewpoints to achieve proper three-dimensional representation.

4 Conclusions

The research presents VSL400 as a publicly available multi-view dataset which serves for word-level VSL recognition of isolated Vietnamese Sign Language (VSL) signs. The dataset contains 74,259 video clips which received manual annotation for 400 glosses. These clips were recorded from 28 signers using three synchronized RGB camera views. The research provides data together with metadata standards and preprocessing methods and evaluation protocols to solve the current lack of Vietnamese Sign Language resources which benefits the entire field of low-resource sign language datasets.

The VSL400 dataset stands apart from other word-level sign language datasets because it includes various linguistic content and uses a specific data collection approach. The current benchmarks AUTSL and CSL-Daily and MS-ASL provide wide-ranging sign language datasets yet they use only single frontal-view video recordings. The system VSL400 operates specifically for word recognition because it unites multiple video streams which record events in perfect synchronization. The design enables researchers to conduct controlled experiments which generate reliable results for studying both how people change their views and how they use three-dimensional signs to express themselves.

The technical validation experiments verified all vital characteristics which the dataset contains. The researcher used signer-disjoint evaluation to verify that VSL400 could recognize new people who it had not encountered before. The data shows the same linguistic and articulatory patterns which replace visual characteristics that belong to specific signers. The development of strong recognition systems for real-world use requires this essential property to work with different types of users.

The evaluation of different camera views shows that using multiple camera angles makes recognition more reliable for all tested model types which include RGB-based and skeleton-based and multimodal systems. The results demonstrate that all modeling approaches produced similar results which indicates that the performance enhancements originated from the data representation methods instead of any specific optimization technique for the architecture. The research findings validate the tri-view acquisition protocol structure because it effectively records the complete three-dimensional process of sign language generation.

The dataset supports multiple representation methods which prevent any preference for working with specific data types. The two models produce results which are simple to understand and match each other because they analyze raw RGB data for appearance-based modeling and skeletal data for pose-based modeling. The ability of VSL400 to work with any input format makes it stand out from other datasets which were created for particular data formats thus enhancing its value as a standard evaluation tool for research studies.

The research results show that VSL400 delivers its best results by using difficult evaluation tests together with its stable measurement system. The reference baselines show per-

formance patterns which follow a predictable pattern while multi-view data enhancements produce positive results for all models tested. The experiments aim to verify the technical aspects of dataset organization and structure instead of performing complete performance assessments.

The VSL400 dataset increases existing sign language databases through its multiple viewing angles which create word-based assessment criteria for a language that needs more representation. The dataset provides an reliable foundation which researchers can use to conduct future studies about isolated and continuous sign language recognition and multi-view and multimodal learning and signer generalization and low-resource modeling. The project makes its data collection available through open access together with its research tools to support research transparency and reproducibility and inclusive studies of Vietnamese Sign Language and visual-gestural communication systems.

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Data Availability

The VSL400 dataset, the preprocessing code used to generate the released data from raw recordings, and the implemented baseline models are publicly available at <https://zenodo.org/uploads/17943574>.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author Contributions Statement

The authors contributed to the study’s conception and design. Material preparation, data collection, and analysis were performed by authors. The first draft of the manuscript was written by Trung Nguyen Quoc; and they commented on previous versions of the manuscript. They read and approved the final manuscript.

Consent to Participate

Informed consent was obtained from all individual participants included in the study.

Consent for Publication

The authors assert that human research subjects granted informed consent for the publishing of all videos. All participants have approved to the submission of the study's results to the journal.

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